

Table 1: Numerical Results for THMRST, RESHYBRID, ADFPRI, MCHCG on Problem 1

#Dim	Ip	THMRST				RESHYBRID				ADFPRI				MCHCG			
		#Iter	#Fev	#Tm	#Nrm	#Iter	#Fev	#Tm	#Nrm	#Iter	#Fev	#Tm	#Nrm	#Iter	#Fev	#Tm	#Nrm
1,000	x1	2	4	0.001351	0	10	28	0.002367	5.61E-06	2	6	0.006612	0	17	19	0.027763	6.23E-08
	x2	1	2	0.001204	0	1	2	0.000644	0	1	4	0.003282	0	14	16	0.013816	4.38E-08
	x3	1	2	0.000787	0	1	2	0.000697	0	1	4	0.001874	0	14	16	0.010708	6.94E-08
	x4	1	4	0.000968	0	9	27	0.001796	4.90E-06	1	4	0.005279	0	13	15	0.008844	9.83E-08
	x5	1	2	0.000659	0	2	4	0.000821	0	1	2	0.004773	0	16	18	0.012037	3.9E-08
5,000	x1	2	4	0.001813	0	10	29	0.006333	8.78E-06	2	6	0.005948	0	18	20	0.030858	4.72E-08
	x2	1	2	0.004435	0	1	2	0.001343	0	1	4	0.002115	0	14	16	0.025062	9.8E-08
	x3	1	2	0.001288	0	1	2	0.001199	0	1	4	0.001469	0	15	17	0.023398	5.25E-08
	x4	1	4	0.001279	0	9	28	0.005845	7.68E-06	1	4	0.001435	0	14	16	0.032488	7.43E-08
	x5	1	2	0.001188	0	2	4	0.001666	0	1	2	0.001582	0	16	18	0.036958	8.73E-08
10,000	x1	2	4	0.003453	0	11	31	0.015555	4.97E-06	2	6	0.004353	0	18	20	0.10073	6.67E-08
	x2	1	2	0.002021	0	1	2	0.002054	0	1	4	0.002817	0	15	17	0.067489	4.69E-08
	x3	1	2	0.002686	0	1	2	0.002207	0	1	4	0.003143	0	15	17	0.0705	7.42E-08
	x4	1	4	0.004125	0	10	30	0.016801	4.35E-06	1	4	0.002671	0	15	17	0.068765	3.56E-08
	x5	1	2	0.003547	0	2	4	0.004126	0	1	2	0.00296	0	17	19	0.065618	4.18E-08
50,000	x1	2	4	0.02466	0	11	32	0.050405	7.78E-06	2	6	0.016723	0	19	21	0.261966	5.04E-08
	x2	1	2	0.00741	0	1	2	0.007472	0	1	4	0.007128	0	16	18	0.258124	3.55E-08
	x3	1	2	0.009821	0	1	2	0.005189	0	1	4	0.006305	0	16	18	0.363361	5.61E-08
	x4	1	4	0.012015	0	10	30	0.048532	9.72E-06	1	4	0.00926	0	15	17	0.235235	7.95E-08
	x5	1	2	0.007397	0	2	4	0.01356	0	1	2	0.00769	0	17	19	0.267343	9.34E-08
100,000	x1	2	4	0.021818	0	12	34	0.16873	4.40E-06	2	6	0.023208	0	19	21	0.664152	7.13E-08
	x2	1	2	0.02205	0	1	2	0.015907	0	1	4	0.01442	0	16	18	0.448895	5.02E-08
	x3	1	2	0.014517	0	1	2	0.009669	0	1	4	0.021877	0	16	18	0.433009	7.94E-08
	x4	1	4	0.013649	0	10	31	0.10205	9.62E-06	1	4	0.016883	0	16	18	0.435603	3.8E-08
	x5	1	2	0.013569	0	2	4	0.02462	0	1	2	0.014332	0	18	20	0.6259	4.47E-08

Table 2: Numerical Results for THMRST, RESHYBRID, ADFPRI, MCHCG on Problem 2

#Dim	Ip	THMRST					RESHYBRID					ADFPRI					MCHCG				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm
1,000	x1	1	2	0.000794	0	1	2	0.00246	0	1	2	0.004311	0	*	*	*	0	*	*	*	*
	x2	1	2	0.000715	0	1	2	0.000761	0	1	4	0.002042	0	*	*	*	0	*	*	*	*
	x3	1	2	0.000727	0	1	2	0.000746	0	1	4	0.001216	0	*	*	*	0	*	*	*	*
	x4	1	4	0.000801	0	1	4	0.000779	0	1	4	0.001546	0	*	*	*	0	*	*	*	*
	x5	1	2	0.000731	0	1	2	0.001442	0	1	2	0.002925	0	*	*	*	0	*	*	*	*
5,000	x1	1	2	0.001199	0	1	2	0.002011	0	1	2	0.002568	0	*	*	*	0	*	*	*	*
	x2	1	2	0.001142	0	1	2	0.001297	0	1	4	0.001922	0	*	*	*	0	*	*	*	*
	x3	1	2	0.001175	0	1	2	0.001229	0	1	4	0.002857	0	*	*	*	0	*	*	*	*
	x4	1	4	0.001452	0	1	4	0.001266	0	1	4	0.001429	0	*	*	*	0	*	*	*	*
	x5	1	2	0.001208	0	1	2	0.001232	0	1	2	0.00117	0	*	*	*	0	*	*	*	*
10,000	x1	1	2	0.001687	0	1	2	0.001795	0	1	2	0.00203	0	*	*	*	0	*	*	*	*
	x2	1	2	0.001844	0	1	2	0.001657	0	1	4	0.002193	0	*	*	*	0	*	*	*	*
	x3	1	2	0.002402	0	1	2	0.002382	0	1	4	0.001887	0	*	*	*	0	*	*	*	*
	x4	1	4	0.001859	0	1	4	0.001886	0	1	4	0.001951	0	*	*	*	0	*	*	*	*
	x5	1	2	0.001518	0	1	2	0.001686	0	1	2	0.002411	0	*	*	*	0	*	*	*	*
50,000	x1	1	2	0.005461	0	1	2	0.004983	0	1	2	0.005173	0	*	*	*	0	*	*	*	*
	x2	1	2	0.006799	0	1	2	0.004661	0	1	4	0.007641	0	*	*	*	0	*	*	*	*
	x3	1	2	0.004779	0	1	2	0.006414	0	1	4	0.009346	0	*	*	*	0	*	*	*	*
	x4	1	4	0.005954	0	1	4	0.008804	0	1	4	0.005584	0	*	*	*	0	*	*	*	*
	x5	1	2	0.004642	0	1	2	0.004669	0	1	2	0.006706	0	*	*	*	0	*	*	*	*
100,000	x1	1	2	0.008434	0	1	2	0.008172	0	1	2	0.009301	0	*	*	*	0	*	*	*	*
	x2	1	2	0.010417	0	1	2	0.009298	0	1	4	0.013923	0	*	*	*	0	*	*	*	*
	x3	1	2	0.008455	0	1	2	0.013742	0	1	4	0.01455	0	*	*	*	0	*	*	*	*
	x4	1	4	0.013228	0	1	4	0.010044	0	1	4	0.014938	0	2	3	0.036727	0	2	3	0.036727	0
	x5	1	2	0.008957	0	1	2	0.012405	0	1	2	0.007856	0	*	*	*	0	*	*	*	*

Table 3: Numerical Results for THMRST, RESHYBRID, ADFPRI, MCHCG on Problem 3

#Dim	Ip	THMRST					RESHYBRID					ADFPRI					MCHCG				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm
1,000	x1	1	5	0.001467	0	1	5	0.000994	0	1	5	0.00825	0	1	2	0.002064	0	1	2	0.002064	0
	x2	1	4	0.00096	0	1	4	0.00132	0	1	4	0.00812	0	1	2	0.00103	0	1	2	0.00103	0
	x3	1	4	0.000862	0	1	4	0.000878	0	1	4	0.00837	0	1	2	0.001177	0	1	2	0.001177	0
	x4	1	4	0.001063	0	1	4	0.000786	0	1	4	0.000746	0	1	2	0.00104	0	1	2	0.00104	0
	x5	1	4	0.001119	0	1	4	0.000787	0	1	4	0.000867	0	1	2	0.00117	0	1	2	0.00117	0
5,000	x1	1	5	0.002279	0	1	5	0.002219	0	1	5	0.00186	0	1	2	0.00232	0	1	2	0.00232	0
	x2	1	4	0.001765	0	1	4	0.001778	0	1	4	0.001693	0	1	2	0.002134	0	1	2	0.002134	0
	x3	1	4	0.002684	0	1	4	0.002209	0	1	4	0.001645	0	1	2	0.001921	0	1	2	0.001921	0
	x4	1	4	0.003537	0	1	4	0.002345	0	1	4	0.001567	0	1	2	0.003647	0	1	2	0.003647	0
	x5	1	4	0.003151	0	1	4	0.001586	0	1	4	0.001641	0	1	2	0.003007	0	1	2	0.003007	0
10,000	x1	1	5	0.003998	0	1	5	0.004437	0	1	5	0.003443	0	1	2	0.004832	0	1	2	0.004832	0
	x2	1	4	0.002777	0	1	4	0.003917	0	1	4	0.004641	0	1	2	0.004548	0	1	2	0.004548	0
	x3	1	4	0.003324	0	1	4	0.003237	0	1	4	0.004093	0	1	2	0.003011	0	1	2	0.003011	0
	x4	1	4	0.004922	0	1	4	0.004338	0	1	4	0.0033	0	1	2	0.003097	0	1	2	0.003097	0
	x5	1	4	0.003964	0	1	4	0.003729	0	1	4	0.003847	0	1	2	0.00344	0	1	2	0.00344	0
50,000	x1	1	5	0.016432	0	1	5	0.017908	0	1	5	0.014714	0	1	2	0.017239	0	1	2	0.017239	0
	x2	1	4	0.013579	0	1	4	0.020052	0	1	4	0.011432	0	1	2	0.012785	0	1	2	0.012785	0
	x3	1	4	0.016571	0	1	4	0.014733	0	1	4	0.013614	0	1	2	0.023848	0	1	2	0.023848	0
	x4	1	4	0.014909	0	1	4	0.021713	0	1	4	0.013418	0	1	2	0.020733	0	1	2	0.020733	0
	x5	1	4	0.012922	0	1	4	0.015762	0	1	4	0.013047	0	1	2	0.015443	0	1	2	0.015443	0
100,000	x1	1	5	0.030903	0	1	5	0.03563	0	1	5	0.026974	0	1	2	0.025976	0	1	2	0.025976	0
	x2	1	4	0.029582	0	1	4	0.044149	0	1	4	0.029909	0	1	2	0.029168	0	1	2	0.029168	0
	x3	1	4	0.036409	0	1	4	0.025968	0	1	4	0.032213	0	1	2	0.034841	0	1	2	0.034841	0
	x4	1	4	0.03114	0	1	4	0.037538	0	1	4	0.026877	0	1	2	0.055629	0	1	2	0.055629	0
	x5	1	4	0.028204	0	1	4	0.031804	0	1	4	0.020496	0	1	2	0.038552	0	1	2	0.038552	0

Table 4: Numerical Results for THMRST, RESHYBRID, ADFPRI, MCHCG on Problem 4

#Dim	Ip	THMRST					RESHYBRID					ADFPRI					MCHCG				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm
1,000	x1	2	9	0.00158	0	1	4	0.000716	0	*	*	*	*	1	2	0.0021	*	1	2	0.0021	0
	x2	2	9	0.001054	0	1	4	0.000835	0	*	*	*	*	1	2	0.001077	*	1	2	0.001077	0
	x3	2	9	0.002106	0	1	4	0.000633	0	*	*	*	*	1	2	0.000901	*	1	2	0.000901	0
	x4	2	9	0.001333	0	1	4	0.00067	0	*	*	*	*	1	2	0.000841	*	1	2	0.000841	0
	x5	2	9	0.001069	0	1	4	0.000676	0	*	*	*	*	1	2	0.001174	*	1	2	0.001174	0
5,000	x1	2	9	0.002219	0	1	4	0.001187	0	*	*	*	*	1	2	0.001947	*	1	2	0.001947	0
	x2	2	9	0.002773	0	1	4	0.002512	0	*	*	*	*	1	2	0.002024	*	1	2	0.002024	0
	x3	2	9	0.002145	0	1	4	0.00172	0	*	*	*	*	1	2	0.002529	*	1	2	0.002529	0
	x4	2	9	0.00368	0	1	4	0.004062	0	*	*	*	*	1	2	0.003475	*	1	2	0.003475	0
	x5	2	9	0.003028	0	1	4	0.001952	0	*	*	*	*	1	2	0.002343	*	1	2	0.002343	0
10,000	x1	2	9	0.006338	0	1	4	0.003183	0	*	*	*	*	1	2	0.003815	*	1	2	0.003815	0
	x2	2	9	0.006265	0	1	4	0.003183	0	*	*	*	*	1	2	0.002361	*	1	2	0.002361	0
	x3	2	9	0.005965	0	1	4	0.001866	0	*	*	*	*	1	2	0.003868	*	1	2	0.003868	0
	x4	2	9	0.005237	0	1	4	0.002005	0	*	*	*	*	1	2	0.003268	*	1	2	0.003268	0
	x5	2	9	0.004591	0	1	4	0.003206	0	*	*	*	*	1	2	0.002362	*	1	2	0.002362	0
50,000	x1	2	9	0.016641	0	1	4	0.008408	0	*	*	*	*	1	2	0.007718	*	1	2	0.007718	0
	x2	2	9	0.018216	0	1	4	0.009259	0	*	*	*	*	1	2	0.020684	*	1	2	0.020684	0
	x3	2	9	0.019235	0	1	4	0.008986	0	*	*	*	*	1	2	0.010544	*	1	2	0.010544	0
	x4	2	9	0.017383	0	1	4	0.009432	0	*	*	*	*	1	2	0.009052	*	1	2	0.009052	0
	x5	2	9	0.023957	0	1	4	0.009317	0	*	*	*	*	1	2	0.008749	*	1	2	0.008749	0
100,000	x1	2	9	0.03914	0	1	4	0.020094	0	*	*	*	*	1	2	0.022106	*	1	2	0.022106	0
	x2	2	9	0.034654	0	1	4	0.016043	0	*	*	*	*	1	2	0.019121	*	1	2	0.019121	0
	x3	2	9	0.035004	0	1	4	0.01731	0	*	*	*	*	1	2	0.019876	*	1	2	0.019876	0
	x4	2	9	0.037332	0	1	4	0.019007	0	*	*	*	*	1	2	0.015699	*	1	2	0.015699	0
	x5	2	9	0.045451	0	1	4	0.014096	0	*	*	*	*	1	2	0.012716	*	1	2	0.012716	0

Table 6: Numerical Results for THMRST, RESHYBRID, ADFPRI, MCHCG on Problem 6

#Dim	Ip	THMRST				RESHYBRID				ADFPRI				MCHCG			
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm
1,000	x1	16	51	0.005859	9.77E-06	22	71	0.008372	9.70E-06	15	49	0.013954	5.25E-06	*	*	*	*
	x2	14	45	0.003605	9.81E-06	24	76	0.008866	8.81E-06	16	52	0.006122	8.73E-06	*	*	*	*
	x3	16	51	0.005555	3.85E-06	24	76	0.007072	8.77E-06	11	37	0.004082	6.39E-06	*	*	*	*
	x4	19	61	0.004713	8.66E-06	22	70	0.005588	9.52E-06	13	43	0.003378	5.96E-06	*	*	*	*
	x5	16	51	0.005836	6.62E-06	18	58	0.007361	8.17E-06	26	84	0.006669	9.52E-06	*	*	*	*
5,000	x1	14	45	0.01448	7.74E-06	24	77	0.0208	9.80E-06	16	53	0.014179	7.95E-06	*	*	*	*
	x2	18	58	0.017784	9.05E-06	25	79	0.018332	9.22E-06	13	43	0.01125	5.69E-06	*	*	*	*
	x3	14	46	0.015312	8.60E-06	17	55	0.014211	7.85E-06	12	41	0.013113	9.46E-06	*	*	*	*
	x4	14	46	0.019398	8.83E-06	25	79	0.019992	9.65E-06	14	47	0.015085	9.61E-06	*	*	*	*
	x5	19	61	0.021385	8.91E-06	20	64	0.017089	9.62E-06	26	83	0.054854	8.91E-06	*	*	*	*
10,000	x1	18	57	0.028962	8.94E-06	20	64	0.04235	8.85E-06	17	55	0.034953	3.88E-06	*	*	*	*
	x2	16	52	0.034474	9.10E-06	26	82	0.051091	8.45E-06	15	49	0.028476	5.37E-06	*	*	*	*
	x3	17	54	0.037822	6.03E-06	19	61	0.038991	7.28E-06	13	43	0.018593	5.19E-06	*	*	*	*
	x4	16	51	0.032218	5.77E-06	19	61	0.030472	9.75E-06	15	49	0.027531	4.87E-06	*	*	*	*
	x5	20	64	0.038398	9.14E-06	21	67	0.042755	8.30E-06	25	81	0.034552	9.77E-06	*	*	*	*
50,000	x1	16	51	0.12278	6.06E-06	20	64	0.16643	8.69E-06	18	58	0.20192	7.71E-06	*	*	*	*
	x2	18	57	0.14141	7.72E-06	25	80	0.1836	9.91E-06	15	49	0.13182	5.33E-06	*	*	*	*
	x3	16	51	0.14924	7.41E-06	20	64	0.14129	8.93E-06	14	47	0.098449	6.93E-06	*	*	*	*
	x4	15	49	0.13805	7.66E-06	27	85	0.18641	9.97E-06	16	52	0.10351	9.99E-06	*	*	*	*
	x5	20	63	0.13979	8.62E-06	18	58	0.13165	9.77E-06	25	80	0.2093	9.02E-06	*	*	*	*
100,000	x1	19	60	0.25098	7.05E-06	20	65	0.32807	9.89E-06	18	59	0.34053	8.41E-06	*	*	*	*
	x2	18	57	0.25946	8.59E-06	18	58	0.27008	7.74E-06	15	49	0.1863	6.75E-06	*	*	*	*
	x3	18	57	0.27502	9.21E-06	17	55	0.26297	7.31E-06	15	49	0.30684	4.66E-06	*	*	*	*
	x4	19	60	0.28576	6.77E-06	22	70	0.34268	8.96E-06	16	53	0.28146	8.85E-06	*	*	*	*
	x5	16	51	0.2485	7.52E-06	17	55	0.27131	9.33E-06	24	78	0.32064	9.87E-06	*	*	*	*

Table 7: Numerical Results for THMRST, RESHYBRID, ADFPRI, MCHCG on Problem 7

#Dim	Ip	THMRST					RESHYBRID					ADFPRI					MCHCG				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm
1,000	x1	38	116	0.024952	7.66E-06	80	243	0.021141	9.33E-06	49	151	0.012931	9.15E-06	160	162	0.100945	9.47E-08				
	x2	27	83	0.013896	4.76E-06	93	282	0.01862	9.47E-06	46	143	0.007277	9.95E-06	104	106	0.056955	8.5E-08				
	x3	23	71	0.004891	6.69E-06	92	279	0.015926	9.65E-06	46	143	0.007433	9.76E-06	124	126	0.06448	9.35E-08				
	x4	32	98	0.006562	7.31E-06	93	283	0.014889	9.93E-06	47	145	0.007138	9.51E-06	94	96	0.048107	9.67E-08				
	x5	27	83	0.010943	9.01E-06	52	159	0.008578	9.60E-06	88	269	0.014149	9.89E-06	157	159	0.079051	9.68E-08				
5,000	x1	32	98	0.020203	5.94E-06	89	270	0.057562	9.96E-06	48	149	0.032009	9.64E-06	139	141	0.36309	8.31E-08				
	x2	22	68	0.013403	9.51E-06	97	294	0.082161	9.53E-06	46	143	0.040051	9.80E-06	159	161	0.498002	9.45E-08				
	x3	25	77	0.014247	3.42E-06	96	291	0.085024	9.68E-06	46	143	0.041444	9.77E-06	220	222	0.564893	8.42E-08				
	x4	29	89	0.016972	4.62E-06	98	297	0.088233	9.73E-06	47	145	0.041598	9.18E-06	195	197	0.465725	9.75E-08				
	x5	38	116	0.023092	4.53E-06	80	243	0.070711	9.87E-06	88	269	0.049892	9.57E-06	131	133	0.283495	8.85E-08				
10,000	x1	33	101	0.039218	9.16E-06	93	282	0.10848	9.50E-06	48	148	0.061567	9.99E-06	147	149	0.873593	9.09E-08				
	x2	36	110	0.054965	8.46E-06	94	285	0.10937	9.90E-06	46	143	0.065024	9.57E-06	216	218	0.985789	9.75E-08				
	x3	31	95	0.041556	4.96E-06	98	297	0.13922	9.52E-06	46	143	0.094844	9.56E-06	208	210	1.00056	9.1E-08				
	x4	40	122	0.066537	7.31E-06	91	277	0.1433	9.93E-06	46	143	0.11978	9.81E-06	338	340	1.702921	8.47E-08				
	x5	28	86	0.046169	7.55E-06	86	261	0.10145	9.38E-06	87	267	0.10428	9.85E-06	243	245	1.169352	8.8E-08				
50,000	x1	29	89	0.21724	4.38E-06	88	267	0.55401	9.48E-06	48	148	0.29129	9.34E-06	286	288	7.124676	9.31E-08				
	x2	36	110	0.2076	8.32E-06	87	264	0.53891	9.58E-06	46	142	0.4139	9.57E-06	240	242	5.799253	9.99E-08				
	x3	33	101	0.28156	6.44E-06	89	270	0.55395	9.97E-06	46	142	0.26814	9.56E-06	79	81	1.850332	8.97E-08				
	x4	68	207	0.38261	9.46E-06	94	285	0.59989	9.40E-06	46	142	0.25748	9.74E-06	109	111	2.493088	9.41E-08				
	x5	35	108	0.28585	9.49E-06	88	267	0.51758	9.45E-06	87	266	0.63772	9.74E-06	334	336	8.258448	9.9E-08				
100,000	x1	28	86	0.33193	4.07E-06	88	267	1.1177	9.29E-06	47	146	0.59625	9.90E-06	420	422	22.84843	8.42E-08				
	x2	68	207	0.91708	9.78E-06	91	276	1.2522	9.56E-06	46	142	0.5697	9.27E-06	361	363	20.03264	9.59E-08				
	x3	41	126	0.4677	9.54E-06	87	264	1.1127	9.67E-06	46	142	0.74137	9.29E-06	317	319	17.17498	8.93E-08				
	x4	28	86	0.34744	4.39E-06	101	306	1.3818	9.46E-06	46	142	0.6301	9.44E-06	284	286	14.04102	8.96E-08				
	x5	35	107	0.4176	8.78E-06	92	279	1.2173	9.43E-06	87	266	1.2766	9.57E-06	479	481	24.63404	8.73E-08				